

SPECIFICATIONS

GENERAL DESCRIPTION

Description	<i>DIFAR, passive, directional</i>
Function	<i>Search, localization, surveillance</i>
Applicable Specification	<i>Production sonobuoy specification dated 26 Oct 94 as modified by the Canadian Forces</i>
Dimensions	<i>36.00 in (914 mm) long by 4.875 in (124 mm) diameter</i>
Weight	<i>16.5 lbs (7.5 kg)</i>
Power Source	<i>Seawater activated battery (main power source) Lithium battery (EFS display and memory)</i>
Stabilization Time (after splash)	<i>Up to 100 seconds (shallow) 160 seconds (medium) 240 seconds (deep)</i>
Operating Life	<i>0.5, 1, 2, 4 or 8 hours preselectable</i>
Scuttling Time	<i>At 8 hours, regardless of operating life</i>
NATO Stock Number	<i>5845-21-909-8480</i>

TRANSMITTER CHARACTERISTICS

Frequency Range	<i>136 MHz to 173.5 MHz</i>
Transmission Channels	<i>99 preselectable, with EFS display</i>
Transmitter RF Power	<i>1 W minimum</i>

SENSOR CHARACTERISTICS

Acoustic Frequency Range	<i>5-2400 Hz</i>
Sensitivity, Directional	<i>122 ± 3 dB rel 1 µPa at 100 Hz = 40kHz pk dev</i>
Sensitivity, Omnidirectional	<i>122 ± 3 dB rel 1 µPa at 100 Hz = 25 kHz pk dev</i>
Operating Depth	<i>30 metres, 120 metres, and 300 metres preselectable</i>
Decent time (in water)	<i>40 seconds (shallow) 100 seconds (medium) 180 seconds (deep)</i>
Directivity, Directional	<i>Horizontal cosine ± 1 dB Vertical cosine ± 3 dB</i>
Directivity, Omnidirectional	<i>± 1 dB horizontal ± 3 dB vertical</i>

OPERATING ENVIRONMENTAL CONDITIONS

Launch Temperature	<i>-20°C to +55°C</i>
Seawater Temperature	<i>0°C to 35°C</i>
Wind Velocity	<i>30 knots (maximum)</i>
Sea State	<i>6 (maximum)</i>
Maximum Launch Altitude	<i>30,000 ft (9144 m)</i>
Maximum Launch Airspeed	<i>370 knots IAS at 1850 ft (564 m)</i>



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AN/SSQ-53D DIFAR

- **Microprocessor-Controlled Full-Function Selector**
 - **Electronic Function Select before launch**
- **Multichannel RF Capability**
- **Multiple Depth and Life Selection Capability**
- **Launchable from Existing ASW Platforms**

The AN/SSQ-53D DIFAR Sonobuoy is the most widely used sensor for all phases of ASW search, localization, and tracking. Basic capabilities include three-depth operation (90, 400 and 1000 feet), five life selections (0.5, 1.0, 2.0, 4.0, and 8.0 hours), DIFAR Sensor, and the capability to set any one of ninety-nine VHF Transmitter channels. These channels may be set via Electronic Function Selection (EFS) prior to launch.

The AN/SSQ-53D can be air launched at airspeeds up to 370 KIAS and altitudes up to 30,000 feet or can be launched from helicopters as low as 40 feet with zero forward velocity.

SPARTON

Passive Sonobuoy

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